FRESH LAB · UBC FACULTY OF FORESTRY & ENVIRONMENTAL STEWARDSHIP

Fire behaviour, with the failures visible.

FRESH Fire Studio is a public benchmark harness for Canadian FBP fire behaviour. A CFS pilot applies the validation ladder to one event or a small benchmark set, with observed and free-burn extent shown separately.

The ask

Bring one event or benchmark set and a technical contact. We will run the full ladder, report where the model diverges, and write a model-limits memo suitable for internal or public release.

What you get

Event evidence

- perimeter and area metrics
- observed daily progression
- free-burn extent and divergence day

Technical report

- validation tier results
- input and provenance record
- failure modes and next tests

What we need

- event perimeter(s), ideally daily
- hourly weather or station/reanalysis source
- fuel and terrain, or agreement to use public layers
- ignition and containment notes if available

Why it is credible

We report published envelopes, not a single pass mark: WildfireAV/ELMFIRE and FARSITE mean Jaccard around 0.18; Prometheus Dogrib 0.21–0.568; FSim BP capture and skill; FOFEM tree mortality. Our own six-case reanalysis benchmark reports median IoU 0.096 and area bias +6.7× to +11.6×. We do not hide that.

What it is not

Not an operational prediction service and not a single-fire calibration exercise. A no-go tier is a valid result.

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